Sham Satish Thakare

Computer Science Researcher & ML Engineer · *Reliable Adaptive Intelligent Systems*

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EDUCATION

APCOER Pune (Anantrao Pawar College of Engineering and Research), Pune, India **01/2020 – 06/2024**

Bachelor of Technology (B.Tech.) in Artificial Intelligence and Data Science | **Cumulative GPA: 8.70 / 10.00**

Core Coursework: Advanced Linear Algebra, Probability & Random Processes, Multivariate Calculus, Discrete Mathematics, Numerical Methods, Machine Learning, Deep Learning, Reinforcement Learning, Distributed Systems, Operating Systems, Database Management Systems, Computer Networks.

RESEARCH EXPERIENCE

Uncertainty-Weighted Credit Assignment in LM RL (Flagship Study) **01/2026 – Present**

- **PROBLEM:** Investigated whether token predictive entropy validly measures reasoning uncertainty in Group Relative Policy Optimization (GRPO) post-training, or operates as a proxy for sequence length.
- **METHOD:** Constructed PyTorch diagnostic probes and a 5-way RL benchmark evaluated across untouched GSM8K mathematical prompts with Qwen2.5-0.5B-Instruct.
- **EVIDENCE:** Uncovered positive correlation between entropy and completion length (r = 0.486, 95% CI [+0.318, +0.627], N=100), which collapsed to non-significance ($r_{partial}$ = -0.092, p = 0.365) after controlling for length; proved nominal MC-dropout was deterministic (Var(log P) = 0.0) on zero-dropout architectures; demonstrated via negative controls that uncertainty weighting produced zero observed online advantage (Cohen's d = 0.00, 80.00% ± 0.00% Pass@1) over standard outcome-supervised GRPO across N=3 seeds.
- **CONTRIBUTION:** Released open-source PyTorch probing suite ear_grpo_reasoning and canonical research essay establishing diagnostic protocols for RLVR credit assignment.

CARLS: Foundation Model RL Plasticity Prediction **02/2026 – Present**

- **PROBLEM:** Addressed computational overhead in executing full policy-gradient RL loops to select optimal pre-trained language model checkpoints for post-training alignment.
- **METHOD:** Formulated representation geometry diagnostic probes analyzing gradient norms and feature drift across Pythia and Qwen checkpoint series.
- **EVIDENCE:** Predicted reward plasticity prior to policy optimization, reducing training FLOPs by 18.7% while preserving post-training Pass@4 accuracy.
- **CONTRIBUTION:** Released open-source RL plasticity benchmarking library adaptive-rl-forge.

MediRush-SafeAgent: Multi-Stage Tool Execution Safety Interception **01/2026 – Present**

- **PROBLEM:** Mitigated unsafe tool execution and policy violation risks in autonomous LLM agents operating over multi-step APIs.
- **METHOD:** Engineered a multi-stage runtime interception architecture combining dynamic pre-execution policy checks, schema validation, and state rollback hooks.
- **EVIDENCE:** Evaluated on healthcare-commerce tool workflows, achieving 100% policy safety interception with 0% unhandled runtime side-effects.
- **CONTRIBUTION:** Developed open-source safety interception framework medirush-safeagent.

QuorumShift: Failure-Domain Aware Consensus Quorum Adaptation **06/2024 – 01/2026**

- **PROBLEM:** Eliminated tail latency spikes and availability degradation in Raft consensus clusters under asymmetric network partitioning.
- **METHOD:** Designed a C++20 dynamic quorum adaptation algorithm that dynamically adjusts voting thresholds and replica selection using real-time latency profiles.
- **EVIDENCE:** Reduced 99th percentile write latency under asymmetric network fault injection while maintaining strict linearizability.
- **CONTRIBUTION:** Built high-performance C++20 open-source library quorumshift.

SELECTED RESEARCH / WORKING PAPERS

• **Thakare, S. S.** "When Confidence Proxies Confound Reasoning Complexity: Pitfalls of Uncertainty-Weighted Credit Assignment in LLMs." *Working Paper / Research Note*, 2026. [PDF Paper] [Code] [Research Note]

• **Thakare, S. S.** "Predicting Reinforcement-Learning Plasticity of Intermediate Language-Model Checkpoints: A Cross-Architecture Diagnostic Study." *Working Paper*, 2026. [Code]

• **Thakare, S. S.** "recovery_eval: State-Matched and Provenance-Aware Evaluation of Recovery Behavior in Language-Model Reasoning." *Working Paper*, 2026. [PDF Paper] [Code]

• **Thakare, S. S.** "AdaptiveReplica: Dynamic Quorum Adaptation and Failure-Aware Replica Selection in Distributed Consensus." *Working Paper*, 2026. [Code]

PROFESSIONAL EXPERIENCE

Spectra Corporate Service, Pune, India **06/2024 – Present**

Data Engineer / Machine Learning Specialist

• Engineered high-throughput parallel data processing pipelines and cloud telemetry infrastructure using C++, Python, and SQL.

• Optimized multithreaded cloud workflow automation scripts and telemetry data ingestion pipelines, reducing task execution latency across compute clusters.

• Designed automated schema verification and error-logging pipelines to maintain data integrity across enterprise data stores.

APCOER Pune (Anantrao Pawar College of Engineering and Research), Pune, India **06/2023 – 05/2024**

Research Intern & Teaching Assistant

• Conducted research on machine learning algorithms and statistical data evaluation methods under department faculty.

• Provided instructional and teaching assistant support for undergraduate coursework in *Data Structures, Algorithms, and Foundations of Machine Learning*, designing lab assignments and grading scripts in Python and C++.

SELECTED OPEN SOURCE ENGINEERING PROJECTS

• **ear_grpo_reasoning (PyTorch Probing Suite)**: Open-source empirical evaluation suite for auditing token predictive entropy, MC-dropout determinism, and advantage scaling in LLM post-training. [GitHub Repository]

• **quorumshift (C++20 Consensus Library)**: High-performance C++20 implementation of dynamic quorum adaptation for Raft consensus under asymmetric network partitioning. [GitHub Repository]

• **medirush-safeagent (Agent Safety Framework)**: Multi-stage runtime interception framework guaranteeing policy safety and execution rollback for autonomous agents. [GitHub Repository]

• **recovery_eval (Reasoning Evaluation Suite)**: State-contrast continuation evaluation framework measuring error recovery under frozen structural covariates. [GitHub Repository]

TECHNICAL SKILLS

• **Programming Languages:** Python, C++ (C++20), SQL, Java, C#/.NET, JavaScript/TypeScript.

• **Machine Learning & AI:** PyTorch, Hugging Face Transformers, GRPO, RLVR, scikit-learn, NumPy, Pandas, Matplotlib, SciPy.

• **Systems & Backend:** C++20 Multithreading, Distributed Consensus (Raft), OpenTelemetry, Microservices, REST APIs, POSIX.

• **Infrastructure & DevOps:** Linux, Git, Docker, GitHub Actions, LaTeX, PyTest, Google Cloud Platform.

• **Research Methods:** Experimental Design, Partial Correlation Analysis, AUROC Evaluation, Multi-Seed Controls, Reproducibility Protocols.